Article

DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning

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General reasoning represents a long-standing and formidable challenge in artificial intelligence (AI). Recent breakthroughs, exemplified by large language models (LLMs)[1,2] and chain-of-thought (CoT) prompting[3], have achieved considerable success on foundational reasoning tasks. However, this success is heavily contingent on extensive human-annotated demonstrations and the capabilities of models are still insufficient for more complex problems. Here we show that the reasoning abilities of LLMs can be incentivized through pure reinforcement learning (RL), obviating the need for human-labelled reasoning trajectories. The proposed RL framework facilitates the emergent development of advanced reasoning patterns, such as self-reflection, verification and dynamic strategy adaptation. Consequently, the trained model achieves superior performance on verifiable tasks such as mathematics, coding competitions and STEM fields, surpassing its counterparts trained through conventional supervised learning on human demonstrations. Moreover, the emergent reasoning patterns exhibited by these large-scale models can be systematically used to guide and enhance the reasoning capabilities of smaller models.

Reasoning capability, the cornerstone of human intelligence, enables complex cognitive tasks ranging from mathematical problem-solving to logical deduction and programming. Recent advances in AI have demonstrated that LLMs can exhibit emergent behaviours, including reasoning abilities, when scaled to a sufficient size[4,5]. However, achieving such capabilities in pre-training typically demands substantial computational resources. In parallel, a complementary line of research has demonstrated that LLMs can be effectively augmented through CoT prompting. This technique, which involves either providing carefully designed few-shot examples or using minimalistic prompts such as "Let's think step by step"[3,6], enables models to produce intermediate reasoning steps, thereby substantially enhancing their performance on complex tasks. Similarly, further performance gains have been observed when models learn high-quality, multistep reasoning trajectories during the post-training phase[2,7]. Despite their effectiveness, these approaches exhibit notable limitations. Their dependence on human-annotated reasoning traces slows scalability and introduces cognitive biases. Furthermore, by constraining models to replicate human thought processes, their performance is inherently capped by the human-provided exemplars, which prevents the exploration of superior, non-human-like reasoning pathways.

To tackle these issues, we aim to explore the potential of LLMs for developing reasoning abilities through self-evolution in a RL framework, with minimal reliance on human labelling efforts. Specifically, we build on DeepSeek-V3 Base[8] and use Group Relative Policy Optimization (GRPO)[9] as our RL framework. The reward signal is only based on the correctness of final predictions against ground-truth answers, without imposing constraints on the reasoning process itself. Notably, we bypass the conventional supervised fine-tuning (SFT) phase before RL training. This design choice originates from our hypothesis that human-defined reasoning patterns may limit model exploration, whereas unrestricted RL training can better incentivize the emergence of new reasoning capabilities in LLMs. Through this process, detailed in the next section, our model (referred to as DeepSeek-R1-Zero) naturally developed diverse and sophisticated reasoning behaviours. To solve reasoning problems, the model exhibits a tendency to generate longer responses, incorporating verification, reflection and the exploration of alternative approaches within each response. Although we do not explicitly teach the model how to reason, it successfully learns improved reasoning strategies through RL.

Although DeepSeek-R1-Zero demonstrates excellent reasoning capabilities, it faces challenges such as poor readability and language mixing, occasionally combining English and Chinese in a single CoT response. Furthermore, the rule-based RL training stage of DeepSeek-R1-Zero is narrowly focused on reasoning tasks, resulting in limited performance in broader areas such as writing and open-domain question answering. To address these challenges, we introduce DeepSeek-R1, a model trained through a multistage learning framework that integrates rejection sampling, RL and supervised fine-tuning, detailed in the 'DeepSeek-R1' section. This training pipeline enables DeepSeek-R1 to inherit the reasoning capabilities of its predecessor, DeepSeek-R1-Zero, while aligning model behaviour with human preferences through further non-reasoning data.

To enable broader access to powerful AI at a lower energy cost, we have distilled several smaller models and made them publicly available. These distilled models exhibit strong reasoning capabilities, surpassing the performance of their original instruction-tuned counterparts. We believe that these instruction-tuned versions will also greatly contribute to the research community by providing a valuable resource for understanding the mechanisms underlying long CoT reasoning models and for promoting the development of more powerful reasoning models. We release DeepSeek-R1-Zero, DeepSeek-R1, data samples and distilled models to the public as described in the 'Code availability' section.

A list of authors and their affiliations appears at the end of the paper.

文章

DeepSeek-R1通过强化学习激励大语言模型的推理能力

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通用推理是人工智能（AI）领域一个长期存在且极为艰巨的挑战。近期的突破，以大语言模型（LLMs）[1,2]和思维链（CoT）提示[3]为代表，在基础推理任务上取得了相当大的成功。然而，这一成功在很大程度上依赖于大量人工标注的演示，且模型的能力对于更复杂的问题仍然不足。在此我们表明，大语言模型的推理能力可以通过纯强化学习（RL）来激励，从而无需人工标注的推理轨迹。所提出的强化学习框架促进了高级推理模式的涌现式发展，例如自我反思、验证和动态策略适应。因此，训练后的模型在数学、编程竞赛和STEM领域等可验证任务上取得了优于通过传统监督学习在人工演示上训练的同类模型的性能。此外，这些大规模模型所展现的涌现推理模式可以被系统地用于指导和增强较小模型的推理能力。

推理能力作为人类智能的基石，使人能够完成从数学问题求解到逻辑推演与编程等复杂的认知任务。近期人工智能的进展表明，当LLMs扩展到足够规模时，可以展现出涌现行为，包括推理能力[4,5]。然而，在预训练阶段获得此类能力通常需要大量计算资源。与此同时，另一条互补的研究路线表明，LLMs可以通过CoT提示得到有效增强。该技术既可以提供精心设计的少样本示例，也可以使用诸如"让我们一步一步思考"这类极简提示[3,6]，使模型生成中间推理步骤，从而显著提升其在复杂任务上的表现。类似地，当模型在后训练阶段学习高质量的多步推理轨迹时，也观察到了进一步的性能提升[2,7]。尽管这些方法有效，但它们存在明显的局限。它们依赖人工标注的推理轨迹，这拖慢了可扩展性并引入认知偏差。此外，通过约束模型去复现人类思维过程，其性能本质上受限于人类提供的示例，从而阻碍了对更优、非人类式推理路径的探索。

为解决这些问题，我们旨在探索LLMs通过在RL框架中自我进化来发展推理能力的潜力，且尽可能减少对人类标注工作的依赖。具体而言，我们基于DeepSeek-V3 Base[8]并使用组相对策略优化（GRPO）[9]作为我们的RL框架。奖励信号仅基于最终预测与真实答案的正确性，而不对推理过程本身施加约束。值得注意的是，我们在RL训练之前绕过了传统的监督微调（SFT）阶段。这一设计选择源于我们的假设：人类定义的推理模式可能会限制模型探索，而无限制的RL训练则能更好地激励LLMs中新推理能力的涌现。通过这一过程（详见下一节），我们的模型（称为DeepSeek-R1-Zero）自然地发展出了多样且复杂的推理行为。为解决推理问题，模型表现出生成更长回答的倾向，在每次回答中纳入验证、反思以及对替代方法的探索。尽管我们没有明确教模型如何推理，它通过RL成功学会了改进的推理策略。

尽管 DeepSeek-R1-Zero 展现出卓越的推理能力，但它面临可读性差和语言混杂等挑战，偶尔在单个 CoT 回复中混合使用英文和中文。此外，DeepSeek-R1-Zero 基于规则的 RL 训练阶段仅狭窄地聚焦于推理任务，导致在写作和开放域问答等更广泛领域的性能有限。为应对这些挑战，我们提出了 DeepSeek-R1，一个通过多阶段学习框架训练的模型，该框架集成了拒绝采样、RL 和监督微调，详见"DeepSeek-R1"章节。这一训练流程使 DeepSeek-R1 能够继承其前身 DeepSeek-R1-Zero 的推理能力，同时通过进一步的非推理数据使模型行为与人类偏好对齐。

为了让更多人能以更低的能耗获得强大的AI，我们蒸馏了多个更小的模型并已将其公开。这些蒸馏模型展现出强大的推理能力，超越了其原始指令微调对应模型的性能。我们相信，这些指令微调版本也将为研究界做出重大贡献，为理解长CoT推理模型的底层机制提供宝贵资源，并推动更强大推理模型的发展。我们按照"代码可用性"部分所述，向公众发布了DeepSeek-R1-Zero、DeepSeek-R1、数据样本和蒸馏模型。

作者列表及其单位信息见文末

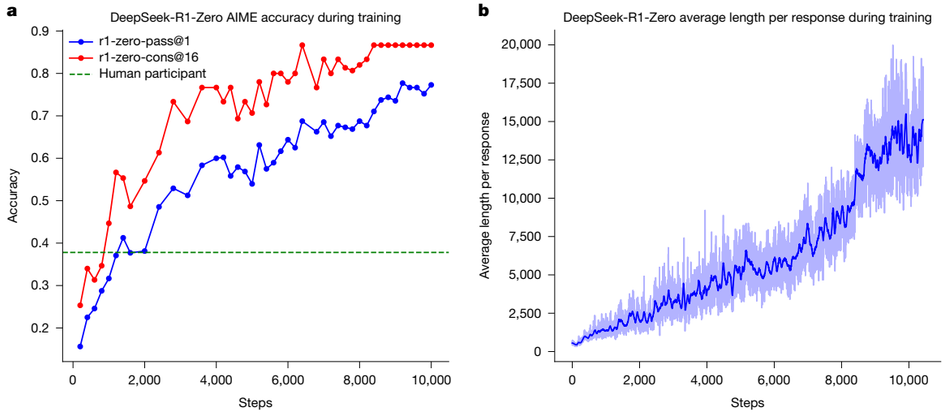


Fig. 1 | Accuracy and output length of DeepSeek-R1-Zero throughout the training process. a, AIME accuracy of DeepSeek-R1-Zero during training. AIME takes a mathematical problem as input and a number as output, illustrated in Extended Data Table 1. pass@1 and cons@16 are described in Supplementary Information, section 4.1. The baseline is the average score achieved by human participants in the AIME competition. **b**, The average response length of DeepSeek-R1-Zero on the training set during the RL process. DeepSeek-R1-Zero naturally learns to solve reasoning tasks with more thinking time. Note that a training step refers to a single policy update operation.

DeepSeek-R1-Zero

To implement large-scale RL of DeepSeek-R1-Zero, we use a highly efficient RL pipeline. Specifically, we use GRPO[9] as our RL algorithm, described in Methods section 'GRPO'. Furthermore, we use a rule-based reward system to compute accuracy and format rewards, with detailed methodologies outlined in Methods section 'Reward design'. Furthermore, our high-performance RL infrastructure is described in Supplementary Information, section 2.1, ensuring scalable and efficient training.

Specifically, we apply the RL technique on the DeepSeek-V3 Base to train DeepSeek-R1-Zero. During training, we design a straightforward template to require DeepSeek-R1-Zero to first produce a reasoning process, followed by the final answer. The prompt template is written as below.

"A conversation between User and Assistant. The User asks a question and the Assistant solves it. The Assistant first thinks about the reasoning process in the mind and then provides the User with the answer. The reasoning process and answer are enclosed within <think>...</think> and <answer>...</answer> tags, respectively, that is, <think> reasoning process here </think><answer> answer here </answer>. User: prompt. Assistant:", in which the prompt is replaced with the specific reasoning question during training. We intentionally limit our constraints to this structural format, avoiding any content-specific biases to ensure that we can accurately observe the natural progression of the model during the RL process.

Figure 1a shows the performance trajectory of DeepSeek-R1-Zero on the American Invitational Mathematics Examination (AIME) 2024 benchmark throughout the RL training process, in which the average pass@1 score on AIME 2024 shows a marked increase, jumping from an initial value of 15.6% to 77.9%. Also, by using the self-consistency decoding[10], the performance of the model can be further improved, achieving an accuracy of 86.7%. This performance greatly surpasses the average performance across all human competitors of the AIME. Besides the maths competitions, as shown in Supplementary Fig. 8, DeepSeek-R1-Zero also achieves remarkable performance in coding competitions and graduate-level biology, physics and chemistry problems. These results underscore the effectiveness of RL in enhancing the reasoning capabilities of LLMs.

As well as the progressive enhancement of reasoning capabilities during training, DeepSeek-R1-Zero also demonstrates self-evolutionary behaviour with RL training. As shown in Fig. 1b, DeepSeek-R1-Zero exhibits a steady increase in thinking time throughout training, driven only by intrinsic adaptation rather than external modifications. Making use of long CoT, the model progressively refines its reasoning, generating hundreds to thousands of tokens to explore and improve its problem-solving strategies.

The increase in thinking time helps with the autonomous development of sophisticated behaviours. Specifically, DeepSeek-R1-Zero increasingly exhibits advanced reasoning strategies such as reflective reasoning and systematic exploration of alternative solutions provided in Extended Data Fig. 1a, substantially boosting its performance on verifiable tasks such as maths and coding. Notably, during training, DeepSeek-R1-Zero exhibits an 'aha moment', shown in Table 1, characterized by a sudden increase in the use of the word 'wait' during reflections, provided in Extended Data Fig. 1b. This moment marks a distinct change in reasoning patterns and clearly shows the self-evolution process of DeepSeek-R1-Zero.

The self-evolution of DeepSeek-R1-Zero underscores the power and beauty of RL: rather than explicitly teaching the model how to solve a problem, we simply provide it with the right incentives and it autonomously develops advanced problem-solving strategies. This serves as a reminder of the potential of RL to unlock higher levels of capabilities in LLMs, paving the way for more autonomous and adaptive models in the future.

DeepSeek-R1

Although DeepSeek-R1-Zero exhibits strong reasoning capabilities, it faces several issues. DeepSeek-R1-Zero struggles with challenges such as poor readability and language mixing, as DeepSeek-V3 Base is trained on several languages, especially English and Chinese. To address these issues, we develop DeepSeek-R1, whose pipeline is illustrated in Fig. 2. In the initial stage, we collect thousands of cold-start data that exhibit a conversational, human-aligned thinking process, as detailed in Supplementary Information, section 2.3.2. RL training is then applied with hyperparameters in Methods section 'Training details of the first RL stage', data details in Supplementary Information, section 2.3.1, to improve the model performance with the

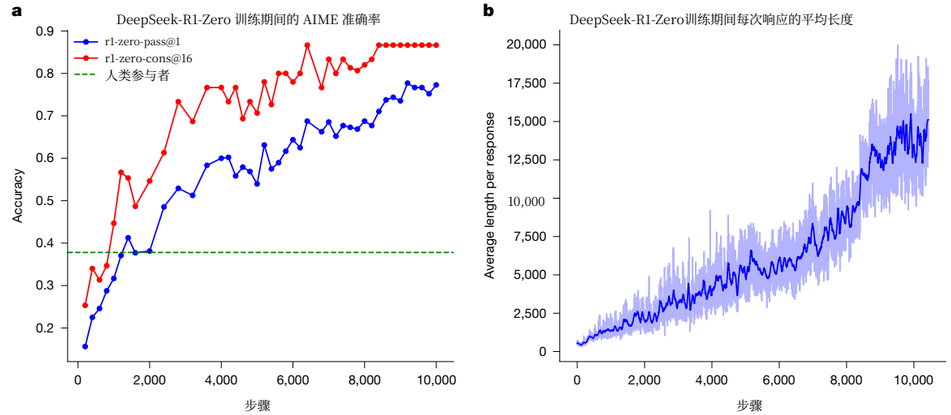


图 1 | DeepSeek-R1-Zero在训练过程中的准确率和输出长度。 a，DeepSeek-R1-Zero在训练期间的AIME准确率。AIME以数学问题为输入，以数字为输出，如扩展数据表1所示。pass@1和cons@16在补充信息第4.1节中描述。基线是人类在AIME竞赛中取得的平均分数。**b**，DeepSeek-R1-Zero在RL过程中在训练集上的平均响应长度。DeepSeek-R1-Zero自然地学会用更多的思考时间来解决推理任务。注意，一个训练步骤指的是单次策略更新操作。

DeepSeek-R1-Zero

为了实现DeepSeek-R1-Zero的大规模强化学习，我们使用了高效的强化学习流水线。具体而言，我们使用GRPO[9]作为我们的强化学习算法，详见方法部分“GRPO”。此外，我们使用基于规则的奖励系统来计算准确性和格式奖励，详细方法见方法部分“奖励设计”。此外，我们的高性能强化学习基础设施见补充信息第2.1节，确保可扩展且高效的训练。

具体而言，我们在DeepSeek-V3 Base上应用强化学习技术来训练DeepSeek-R1-Zero。在训练过程中，我们设计了一个简单的模板，要求DeepSeek-R1-Zero首先生成推理过程，然后给出最终答案。提示模板如下所示。

“用户与助手之间的对话。用户提出一个问题，助手解决它。助手首先在脑海中思考推理过程，然后向用户提供答案。推理过程和答案分别被包含在<think>...</think>and <answer>...</answer> tags中，即<think> reasoning在此处放置过程</think><answer> answer在此处放置</answer>.用户：提示。助手：”，在训练中将提示替换为具体的推理问题。我们有意将约束限制在这种结构格式上，避免任何内容特定的偏差，以确保我们能够准确观察模型在强化学习过程中的自然进展。

图 **1a**展示了DeepSeek-R1-Zero在美国数学邀请赛（AIME）2024基准测试中，在整个强化学习训练过程中的性能轨迹。其中，AIME 2024上的平均pass@1得分显著提升，从初始的15.6%跃升至77.9%。此外，通过使用自一致性解码[10]，模型的性能可进一步提升，达到86.7%的准确率。这一表现远超AIME所有人类参赛者的平均成绩。除数学竞赛外，如补充图8所示，DeepSeek-R1-Zero在编程竞赛以及研究生级别的生物、物理和化学问题上也表现出色。这些结果凸显了强化学习在提升大语言模型推理能力方面的有效性。

除了训练过程中推理能力的逐步增强外，DeepSeek-R1-Zero在强化学习训练中还展现出自我进化行为。如图**1b**所示，DeepSeek-R1-Zero在整个训练过程中思考时间稳步增加，这仅由内在适应驱动，而非外部修改。利用长思维链，模型逐步完善其推理过程，生成数百至数千个token以探索和改进其问题解决策略。

思考时间的增加有助于自主发展复杂行为。具体而言，DeepSeek-R1-Zero 日益展现出如反思性推理和对备选解决方案的系统性探索等高级推理策略（见扩展数据图 1a），显著提升了其在数学和编码等可验证任务上的表现。值得注意的是，在训练过程中，DeepSeek-R1-Zero 表现出一个‘顿悟时刻’，见表 **1**，其特征是在反思过程中‘wait’一词的使用突然增加（见扩展数据图 1b）。这一时刻标志着推理模式的明显变化，并清晰展示了 DeepSeek-R1-Zero 的自我演化过程。

DeepSeek-R1-Zero 的自我演化凸显了强化学习的力量与美妙之处：我们并非明确教给模型如何解决问题，而只是为其提供正确的激励，它便自主发展出高级的问题解决策略。这提醒我们，强化学习有潜力释放大语言模型更高层次的能力，为未来更自主、更具适应性的模型铺平道路。

DeepSeek-R1

尽管DeepSeek-R1-Zero展现出强大的推理能力，但它面临若干问题。DeepSeek-R1-Zero在可读性差和语言混杂等方面存在困难，因为DeepSeek-V3 Base是在多种语言上训练的，尤其是英语和中文。为了解决这些问题，我们开发了DeepSeek-R1，其流程如图**2**所示。在初始阶段，我们收集了数千条冷启动数据，这些数据展现出对话式、符合人类对齐的思维过程，详见补充信息第2.3.2节。随后应用RL训练，超参数见方法部分“第一RL阶段的训练细节”，数据细节见补充信息第2.3.1节，以提升模型在

Table 1 | An interesting 'aha moment' of an intermediate version of DeepSeek-R1-Zero

Question: If $a>1$, then the sum of the real solutions of $\sqrt{a-\sqrt{a+x}}=x$ is equal to

Response: <think>

To solve the equation $\sqrt{a-\sqrt{a+x}}=x$, let's start by squaring both ...

$(\sqrt{a-\sqrt{a+x}})^2=x^2 \Rightarrow a-\sqrt{a+x}=x^2$.

Rearrange to isolate the inner square root term:

$(a-x^2)^2=a+x \Rightarrow a^2-2ax^2+(x^2)^2=a+x \Rightarrow x^4-2ax^2-x+(a^2-a)=0$

...

Wait, wait. Wait. That's an aha moment I can flag here.

Let's reevaluate this step by step to identify whether the correct sum can be ...

We started with the equation:

$\sqrt{a-\sqrt{a+x}}=x$

First, let's square both sides:

$a-\sqrt{a+x}=x^2 \Rightarrow \sqrt{a+x}=a-x^2$

Next, I could square both sides again, treating the equation: ...

...

The model learns to rethink using an anthropomorphic tone. This is also an aha moment for us, allowing us to witness the power and beauty of RL.

conversational thinking process and language consistency. Subsequently, we apply rejection sampling and SFT once more. This stage incorporates both reasoning and non-reasoning datasets into the SFT process, as detailed in Supplementary Information, section 2.3.3, enabling the model to not only excel in reasoning tasks but also demonstrate advanced writing capabilities. To further align the model with human preferences, we implement a secondary RL stage designed to enhance the helpfulness and harmlessness of the model while simultaneously refining its reasoning capabilities. The reward model is described in Methods section 'Reward design' and RL hyperparameters are in Methods section 'Training details of the second RL stage'. The total training cost is listed in Supplementary Information, section 2.4.4.

We evaluate our models on MMLU[11], MMLU-Redux[12], MMLU-Pro[13], DROP[14], C-Eval[15], IFEval[16], FRAMES[17], GPQA Diamond[18], SimpleQA[19], C-SimpleQA[20], CLUEWSC[21], AlpacaEval 2.0 (ref. 22), Arena-Hard[23], SWE-bench Verified[24], Aider-Polyglot[25], LiveCodeBench[26] (2024-08–2025-01), Codeforces[27], Chinese National High School Mathematics Olympiad (CNMO 2024)[28] and AIME 2024 (ref. 29). The details of these benchmarks are provided in Supplementary Tables 15–29.

Table 2 summarizes the performance of DeepSeek-R1 across several developmental stages, as outlined in Fig. 2. A comparison between DeepSeek-R1-Zero and DeepSeek-R1 Dev1 reveals substantial improvements in instruction-following, as evidenced by higher scores on the IF-Eval and Arena-Hard benchmarks. However, owing to the limited size of the cold-start dataset, Dev1 exhibits a partial degradation in reasoning performance compared with DeepSeek-R1-Zero, most notably on the AIME benchmark. By contrast, DeepSeek-R1 Dev2 demonstrates marked performance enhancements on benchmarks that require advanced reasoning skills, including those focused on code generation, mathematical problem solving and STEM-related tasks. Benchmarks targeting general-purpose tasks, such as Alpac-aEval 2.0, show marginal improvement. These results indicate that reasoning-oriented RL considerably enhances reasoning capabilities while exerting limited influence on user-preference-oriented benchmarks.

DeepSeek-R1 Dev3 integrates both reasoning and non-reasoning datasets into the SFT pipeline, thereby enhancing the proficiency of the model in both reasoning and general language-generation tasks. Compared with Dev2, DeepSeek-R1 Dev3 achieves notable performance improvements on AlpacaEval 2.0 and Aider-Polyglot, attributable to the inclusion of large-scale non-reasoning corpora and code-engineering datasets. Finally, comprehensive RL training on DeepSeek-R1 Dev3 using mixed reasoning-focused and general-purpose data produced the final DeepSeek-R1. Marginal improvements occurred in code and mathematics benchmarks, as substantial reasoning-specific RL was done in previous stages. The primary advancements in the final DeepSeek-R1 were in general instruction-following and user-preference benchmarks, with AlpacaEval 2.0 improving by 25% and Arena-Hard by 17%.

We also compare DeepSeek-R1 with other models in Supplementary Information, section 4.2. Model safety evaluations are provided in Supplementary Information, section 4.3. A comprehensive analysis of evaluation is provided in Supplementary Information, section 5, including a comparison with DeepSeek-V3, performance evaluations on both fresh test sets, a breakdown of mathematical capabilities by category and an investigation of test-time scaling behaviour. Supplementary Information, section 6 shows that the strong reasoning capability can be transferred to smaller models.

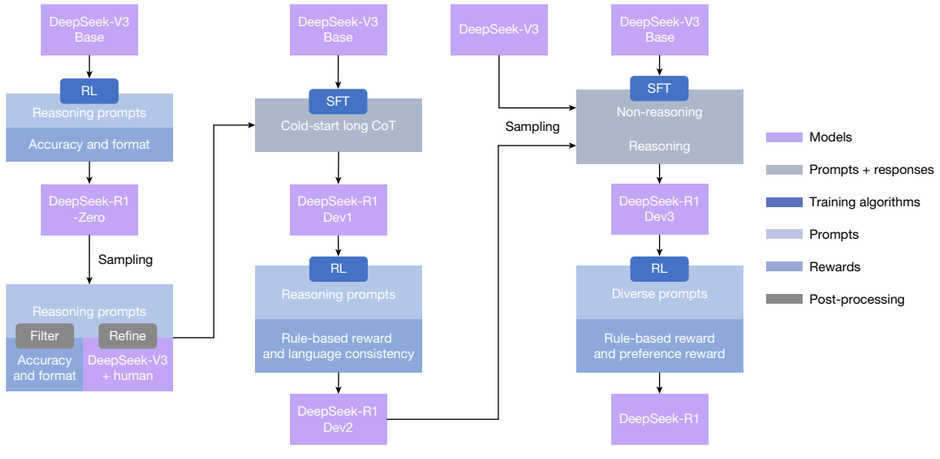


Fig. 2 | The multistage pipeline of DeepSeek-R1. A detailed background on DeepSeek-V3 Base and DeepSeek-V3 is provided in Supplementary Information, section 1.1. The models DeepSeek-R1 Dev1, Dev2 and Dev3 represent intermediate checkpoints in this pipeline.

表 1 | 一个有趣的“顿悟时刻”，来自中间版本的 DeepSeek-R1-Zero

问题：如果 $a>1$，那么 a 的实数解之和为 $\sqrt{a-\sqrt{a+x}}=x$ 等于

响应：<think>

为了求解方程 a $\sqrt{a-\sqrt{a+x}}=x$，让我们先对两边进行平方……

$^2\sqrt{^2-\sqrt{^2+x}}=x^2 \Rightarrow a-\sqrt{a+x}=x^2$.

重新排列以分离内层平方根项：

$(a-x^2)^2=a+x \Rightarrow a^2-2ax^2+(x^2)^2=a+x \Rightarrow x^4-2ax^2-x+(a^2-a)=0$

...

等等，等等。等等。这是一个我可以在这里标记的顿悟时刻。

让我们逐步重新评估这一步，以确定正确的和是否可能为……

我们从以下方程开始：

$\sqrt{a-\sqrt{a+x}}=x$

首先，让我们对两边进行平方：

$a-\sqrt{a+x}=x^2 \Rightarrow \sqrt{a+x}=a-x^2$

接下来，我可以再次对两边进行平方，处理该方程：……

...

该模型学会了用拟人化的语气重新思考。这也是一个顿悟时刻，对于我们，让我们得以见证强化学习的力量与美妙。

对话式思维过程和语言一致性方面的性能。随后，我们再次应用拒绝采样和SFT。此阶段将推理和非推理数据集都纳入SFT过程，详见补充信息第2.3.3节，使模型不仅在推理任务上表现出色，还能展现高级写作能力。为了进一步使模型与人类偏好对齐，我们实施了第二RL阶段，旨在增强模型的有用性和无害性，同时改进其推理能力。奖励模型见方法部分“奖励设计”，RL超参数见方法部分“第二RL阶段的训练细节”。总训练成本列于补充信息第2.4.4节。

我们在 MMLU[11], MMLU-Redux[12], MMLU-Pro[13],DROP[14], C-Eval[15], IFEval[16], FRAMES[17], GPQA Diamond[18], SimpleQA[19],C-SimpleQA[20], CLUEWSC[21], AlpacaEval 2.0（参考文献 22）、Arena-Hard[23],SWE-bench Verified[24], Aider-Polyglot[25], LiveCodeBench[26]（2024-08– 2025-01）、Codeforces[27], 中国全国高中数学奥林匹克竞赛（CNMO 2024）[28] 和 AIME 2024（参考文献 29）。这些基准测试的详细信息见补充表 15–29。

表 2 总结了 DeepSeek-R1 在多个开发阶段的性能，如图 2 所示。DeepSeek-R1-Zero 与 DeepSeek-R1 Dev1 的对比显示，指令遵循能力有显著提升，这体现在 IF-Eval 和 Arena-Hard 基准测试得分更高上。然而，由于冷启动数据集规模有限，与 DeepSeek-R1-Zero 相比，Dev1 在推理性能上出现了部分退化，在 AIME 基准测试上尤为明显。相比之下，DeepSeek-R1 Dev2 在需要高级推理技能的基准测试上表现出显著性能提升，包括代码生成、数学问题求解和 STEM 相关任务。面向通用任务的基准测试，如 Alpac-aEval 2.0，仅显示出 marginal improvement。这些结果表明，面向推理的强化学习大幅增强了推理能力，而对面向用户偏好的基准测试影响有限。

DeepSeek-R1 Dev3 将推理数据集和非推理数据集整合到 SFT 流程中，从而增强了模型在推理和通用语言生成任务上的能力。与 Dev2 相比，DeepSeek-R1 Dev3 在 AlpacaEval 2.0 和 Aider-Polyglot 上取得了显著的性能提升，这归因于引入了大规模非推理语料库和代码工程数据集。最后，使用混合的推理导向和通用数据对 DeepSeek-R1 Dev3 进行全面的 RL 训练，产生了最终的 DeepSeek-R1。在代码和数学基准测试中出现了边际改进，因为在之前的阶段已经进行了大量针对推理的 RL。最终 DeepSeek-R1 的主要进步在于通用指令遵循和用户偏好基准测试，其中 AlpacaEval 2.0 提升了 25%，Arena-Hard 提升了 17%。

我们还在补充信息第4.2节中将DeepSeek-R1与其他模型进行了比较。模型安全性评估见补充信息第4.3节。评估的全面分析见补充信息第5节，包括与DeepSeek-V3的比较、在两个全新测试集上的性能评估、按类别划分的数学能力分析以及测试时扩展行为的研究。补充信息第6节表明，强大的推理能力可以迁移到更小的模型上。

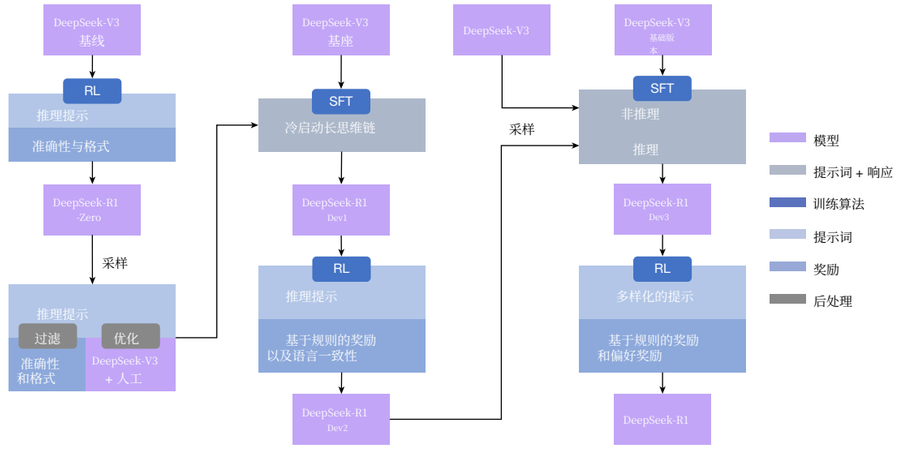


图 2 | DeepSeek-R1 的多阶段流程。有关 DeepSeek-V3 Base 和 DeepSeek-V3 的详细背景见补充信息第 1.1 节。模型 DeepSeek-R1 Dev1、Dev2 和 Dev3 表示该流程中的中间检查点。

Table 2 | Experimental results at each stage of DeepSeek-R1

	Benchmark (metric)	R1-Zero	R1Dev1	R1Dev2	R1Dev3	R1
English	MMLU (EM)	88.8	89.1	**91.2**	91.0	90.8
	MMLU-Redux (EM)	85.6	90.0	93.0	93.1	92.9
	MMLU-Pro (EM)	68.9	74.1	83.8	83.1	**84.0**
	DROP (3-shot F1)	89.1	89.8	91.1	88.7	**92.2**
	IF-Eval (Prompt Strict)	46.6	71.7	72.0	78.1	**83.3**
	GPQA Diamond (Pass@1)	**75.8**	66.1	70.7	71.2	71.5
	SimpleQA (Correct)	30.3	17.8	28.2	24.9	30.1
	FRAMES (Acc.)	82.3	78.5	81.8	81.9	**82.5**
	AlpacaEval2.0 (LC-winrate)	24.7	50.1	55.8	62.1	**87.6**
	Arena-Hard (GPT-4-1106)	53.6	77.0	73.2	75.6	**92.3**
Code	LiveCodeBench (Pass@1-COT)	50.0	57.5	63.5	64.6	**65.9**
	Codeforces (Percentile)	80.4	84.5	90.5	92.1	**96.3**
	Codeforces (Rating)	1,444	1,534	1,687	1,746	**2,029**
	SWE-bench Verified (Resolved)	43.2	39.6	44.6	45.6	**49.2**
	Aider-Polyglot (Acc.)	12.2	6.7	25.6	44.8	**53.3**
Maths	AIME2024 (Pass@1)	77.9	59.0	74.0	78.1	**79.8**
	MATH-500 (Pass@1)	95.9	94.2	95.9	95.4	**97.3**
	CNMO2024 (Pass@1)	**88.1**	58.0	73.9	77.3	78.8
Chinese	CLUEWSC (EM)	93.1	92.8	92.6	91.6	92.8
	C-Eval (EM)	**92.8**	85.7	91.9	86.4	91.8
	C-SimpleQA (Correct)	66.4	58.8	64.2	66.9	63.7

Numbers in bold denote that the performance is statistically significant (*t*-test with $P<0.01$).

Ethics and safety statement

With the advancement in the reasoning capabilities of DeepSeek-R1, we deeply recognize the potential ethical risks. For example, R1 can be subject to jailbreak attacks, leading to the generation of dangerous content such as explosive manufacturing plans, whereas the enhanced reasoning capabilities enable the model to provide plans with better operational feasibility and executability. Besides, a public model is also vulnerable to further fine-tuning that could compromise inherent safety protections.

In Supplementary Information, section 4.3, we present a comprehensive safety report from several perspectives, including performance on open-source and in-house safety evaluation benchmarks, and safety levels across several languages and against jailbreak attacks. These comprehensive safety analyses conclude that the inherent safety level of the DeepSeek-R1 model, compared with other state-of-the-art models, is generally at a moderate level (comparable with GPT-4o (2024-05-13)[30]). Besides, when coupled with the risk control system, the safety level of the model is increased to a superior standard.

Conclusion, limitation and future work

We present DeepSeek-R1-Zero and DeepSeek-R1, which rely on large-scale RL to incentivize model reasoning behaviours. Our results demonstrate that pre-trained checkpoints inherently have substantial potential for complex reasoning tasks. We believe that the key to unlocking this potential lies not in large-scale human annotation but in the provision of hard reasoning questions, a reliable verifier and sufficient computational resources for RL. Sophisticated reasoning behaviours, such as self-verification and reflection, seemed to emerge organically during the RL process.

Even if DeepSeek-R1 achieves frontier results on reasoning benchmarks, it still faces several capability limitations, as outlined below.

Structure output and tool use

At present, the structural output capabilities of DeepSeek-R1 remain suboptimal compared with existing models. Moreover, DeepSeek-R1 cannot make use of tools, such as search engines and calculators, to improve the performance of output. However, as it is not hard to build a RL environment for structure output and tool use, we believe that the issue will be addressed in the next version.

Token efficiency

Unlike conventional test-time computation scaling approaches, such as majority voting or Monte Carlo tree search (MCTS), DeepSeek-R1 dynamically allocates computational resources during inference according to the complexity of the problem at hand. Specifically, it uses fewer tokens to solve simple tasks but generating more tokens for complex tasks. Nevertheless, there remains room for further optimization in terms of token efficiency, as instances of excessive reasoning—manifested as overthinking—are still observed in response to simpler questions.

Language mixing

DeepSeek-R1 is at present optimized for Chinese and English, which may result in language-mixing issues when handling queries in other languages. For instance, DeepSeek-R1 might use English for reasoning and responses, even if the query is in a language other than English or Chinese. We aim to address this limitation in future updates. The limitation may be related to the base checkpoint, DeepSeek-V3 Base, which mainly uses Chinese and English, so that it can achieve better results with the two languages in reasoning.

Prompting engineering

When evaluating DeepSeek-R1, we observe that it is sensitive to prompts. Few-shot prompting consistently degrades its performance. Therefore, we recommend that users directly describe the problem and specify the output format using a zero-shot setting for optimal results.

表 2 | DeepSeek-R1 各阶段的实验结果

	基准测试（指标）	R1-Zero	R1 Dev1	R1 Dev2	R1 Dev3	R1
英语	MMLU (EM)	88.8	89.1	**91.2**	91.0	90.8
	MMLU-Redux (EM)	85.6	90.0	93.0	93.1	92.9
	MMLU-Pro (EM)	68.9	74.1	83.8	83.1	**84.0**
	DROP (3-shot F1)	89.1	89.8	91.1	88.7	**92.2**
	IF-Eval（提示严格）	46.6	71.7	72.0	78.1	**83.3**
	GPQA Diamond（Pass@1）	**75.8**	66.1	70.7	71.2	71.5
	SimpleQA（正确）	30.3	17.8	28.2	24.9	30.1
	FRAMES（准确率）	82.3	78.5	81.8	81.9	**82.5**
	AlpacaEval 2.0（LC胜率）	24.7	50.1	55.8	62.1	**87.6**
	Arena-Hard（GPT-4-1106）	53.6	77.0	73.2	75.6	**92.3**
代码	LiveCodeBench（Pass@1-COT）	50.0	57.5	63.5	64.6	**65.9**
	Codeforces（百分位）	80.4	84.5	90.5	92.1	**96.3**
	Codeforces（评分）	1,444	1,534	1,687	1,746	**2,029**
	SWE-bench Verified（已解决）	43.2	39.6	44.6	45.6	**49.2**
	Aider-Polyglot（准确率）	12.2	6.7	25.6	44.8	**53.3**
数学	AIME 2024（Pass@1）	77.9	59.0	74.0	78.1	**79.8**
	MATH-500（Pass@1）	95.9	94.2	95.9	95.4	**97.3**
	CNMO 2024（Pass@1）	**88.1**	58.0	73.9	77.3	78.8
中文	CLUEWSC（EM）	93.1	92.8	92.6	91.6	92.8
	C-Eval（EM）	**92.8**	85.7	91.9	86.4	91.8
	C-SimpleQA（正确）	66.4	58.8	64.2	66.9	63.7

粗体数字表示性能具有统计显著性（*t*检验，$P<0.01$）。

伦理与安全声明

随着DeepSeek-R1推理能力的进步，我们深刻认识到潜在的伦理风险。例如，R1可能遭受越狱攻击，导致生成危险内容，如爆炸物制造方案，而增强的推理能力使模型能够提供具有更好操作可行性和可执行性的方案。此外，公开模型也容易受到进一步微调的影响，这可能破坏固有的安全保护。

在补充信息第4.3节中，我们从多个角度呈现了一份全面的安全报告，包括在开源及内部安全评估基准上的表现，以及跨多种语言和抵御越狱攻击的安全水平。这些全面的安全分析得出结论：与其他最先进模型相比，DeepSeek-R1模型的固有安全水平总体处于中等水平（与GPT-4o (2024-05-13)相当）[30]。此外，当与风险控制系统结合时，该模型的安全水平提升至更优标准。

结论、局限性与未来工作

我们提出了DeepSeek-R1-Zero和DeepSeek-R1，它们依靠大规模强化学习来激励模型的推理行为。我们的结果表明，预训练检查点本身在复杂推理任务上具有巨大潜力。我们相信，释放这一潜力的关键不在于大规模人工标注，而在于提供困难的推理问题、一个可靠的验证器以及足够的强化学习计算资源。诸如自我验证和反思等复杂的推理行为，似乎在强化学习过程中自然涌现。

即使 DeepSeek-R1 在推理基准测试中取得了前沿结果，它仍然面临若干能力限制，概述如下。

结构化输出与工具使用

目前，与现有模型相比，DeepSeek-R1 的结构化输出能力仍不够理想。此外，DeepSeek-R1 无法利用搜索引擎、计算器等工具来提升输出性能。然而，由于构建用于结构化输出和工具使用的 RL 环境并不困难，我们相信该问题将在下一个版本中得到解决。

令牌效率

与传统的测试时计算扩展方法（如多数投票或蒙特卡洛树搜索（MCTS））不同，DeepSeek-R1 在推理过程中根据当前问题的复杂度动态分配计算资源。具体而言，它使用较少的令牌来解决简单任务，但为复杂任务生成更多令牌。尽管如此，在令牌效率方面仍有进一步优化的空间，因为在回答较简单问题时，仍可观察到过度推理的实例——表现为过度思考。

语言混合

DeepSeek-R1 目前针对中文和英文进行了优化，这可能导致在处理其他语言的查询时出现语言混合问题。例如，即使查询使用的是英语或中文之外的语言，DeepSeek-R1 也可能使用英语进行推理和回复。我们计划在未来的更新中解决这一限制。该限制可能与基础检查点 DeepSeek-V3 Base 有关，它主要使用中文和英文，因此能够在这两种语言的推理中取得更好的结果。

提示工程

在评估DeepSeek-R1时，我们观察到它对提示敏感。少样本提示会持续降低其性能。因此，我们建议用户直接描述问题并以零样本设置指定输出格式，以获得最佳结果。

Software-engineering tasks

Owing to the long evaluation times, which affect the efficiency of the RL process, large-scale RL has not been applied extensively in software-engineering tasks. As a result, DeepSeek-R1 has not demonstrated a huge improvement over DeepSeek-V3 on software-engineering benchmarks. Future versions will address this by implementing rejection sampling on software-engineering data or incorporating asynchronous evaluations during the RL process to improve efficiency.

Beyond specific capability limitations, the pure RL methodology itself also presents inherent challenges:

Reward hacking

The success of pure RL depends on reliable reward signals. In this study, we ensure reward reliability through a reasoning-domain rule-based reward model. However, such dependable reward models are difficult to construct for certain tasks, such as writing. If the reward signal is assigned by a model instead of predefined rules, it becomes more susceptible to exploitation as training progresses, which means that the policy model may find shortcuts to hack the reward model. Consequently, for complex tasks that cannot be effectively evaluated by a reliable reward model, scaling up pure RL methods remains an open challenge.

In this work, for tasks that cannot obtain a reliable signal, DeepSeek-R1 uses human annotation to create supervised data and only conducts RL for hundreds of steps. We hope that, in the future, a robust reward model can be obtained to address such issues.

With the advent of pure RL methods such as DeepSeek-R1, the future holds immense potential for solving any task that can be effectively evaluated by a verifier, regardless of its complexity for humans. Machines equipped with such advanced RL techniques are poised to surpass human capabilities in these domains, driven by their ability to optimize performance iteratively through trial and error. However, challenges remain for tasks for which constructing a reliable reward model is inherently difficult. In such cases, the lack of a robust feedback mechanism may slow progress, suggesting that future research should focus on developing innovative approaches to define and refine reward structures for these complex, less verifiable problems.

Furthermore, making use of tools during the reasoning process holds notable promise. Whether it is using tools such as compilers or search engines to retrieve or compute necessary information or using external tools such as biological or chemical reagents to validate final results in the real world, this integration of tool-augmented reasoning could greatly enhance the scope and accuracy of machine-driven solutions.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-025-09422-z.

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软件工程任务

由于评估时间较长，这影响了强化学习过程的效率，大规模强化学习尚未在软件工程任务中广泛应用。因此，DeepSeek-R1在软件工程基准上未表现出相较于DeepSeek-V3的巨大改进。未来版本将通过实施软件工程数据上的拒绝采样或在强化学习过程中引入异步评估来提高效率。

除了特定的能力局限外，纯强化学习方法本身也存在固有挑战：

奖励黑客

纯强化学习的成功依赖于可靠的奖励信号。在本研究中，我们通过推理领域的基于规则的奖励模型来确保奖励的可靠性。然而，对于某些任务（如写作），构建如此可靠的奖励模型十分困难。如果奖励信号由模型而非预定义规则给出，那么随着训练的推进，它更容易被利用，这意味着策略模型可能找到捷径来攻击奖励模型。因此，对于无法通过可靠奖励模型有效评估的复杂任务，扩展纯强化学习方法仍是一个未解决的挑战。

在本工作中，对于无法获得可靠信号的任务，DeepSeek-R1使用人工标注来创建监督数据，并且仅进行数百步的强化学习。我们希望未来能够获得一个稳健的奖励模型来解决此类问题。

随着DeepSeek-R1等纯强化学习方法的出现，未来在解决任何能够被验证器有效评估的任务方面蕴含着巨大潜力，无论这些任务对人类而言多么复杂。配备此类先进强化学习技术的机器有望在这些领域超越人类能力，这得益于其通过试错迭代优化性能的能力。然而，对于本质上难以构建可靠奖励模型的任务，挑战依然存在。在这种情况下，缺乏稳健的反馈机制可能会减缓进展，这表明未来的研究应聚焦于开发创新方法，为这些复杂且较难验证的问题定义并改进奖励结构。

此外，在推理过程中利用工具具有显著的前景。无论是使用编译器或搜索引擎等工具来检索或计算必要信息，还是使用生物或化学试剂等外部工具在现实世界中验证最终结果，这种工具增强推理的整合都能极大提升机器驱动解决方案的范围和准确性。

在线内容

任何方法、附加参考文献、Nature Portfolio报告摘要、源数据、扩展数据、补充信息、致谢、同行评审信息；作者贡献和利益冲突详情；以及数据和代码可用性声明均可在以下网址获取：https://doi.org/10.1038/s41586-025-09422-z。

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Methods

GRPO

GRPO[9] is the RL algorithm that we use to train DeepSeek-R1-Zero and DeepSeek-R1. It was originally proposed to simplify the training process and reduce the resource consumption of proximal policy optimization (PPO)[31], which is widely used in the RL stage of LLMs[32]. The pipeline of GRPO is shown in Extended Data Fig. 2.

For each question q, GRPO samples a group of outputs $\{o_1, o_2, \ldots, o_G\}$ from the old policy $\pi_{\theta_{\text{old}}}$ and then optimizes the policy model π_θ by maximizing the following objective:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}[q \sim P(Q), \{o_i\}_{i=1}^{G} \sim \pi_{\theta_{\text{old}}}(O|q)]$$
$$\frac{1}{G}\sum_{i=1}^{G}\left(\min\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{\text{old}}}(o_i|q)}A_i,\ \text{clip}\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{\text{old}}}(o_i|q)}, 1-\epsilon, 1+\epsilon\right)A_i\right)\right. \quad (1)$$
$$\left. - \beta\mathbb{D}_{KL}(\pi_\theta||\pi_{\text{ref}})\right),$$

$$\mathbb{D}_{\text{KL}}(\pi_\theta||\pi_{\text{ref}}) = \frac{\pi_{\text{ref}}(o_i|q)}{\pi_\theta(o_i|q)} - \log\frac{\pi_{\text{ref}}(o_i|q)}{\pi_\theta(o_i|q)} - 1, \quad (2)$$

in which π_{ref} is a reference policy, ϵ and β are hyperparameters and A_i is the advantage, computed using a group of rewards $\{r_1, r_2, \ldots, r_G\}$ corresponding to the outputs in each group:

$$A_i = \frac{r_i - \text{mean}(\{r_1, r_2, \cdots, r_G\})}{\text{std}(\{r_1, r_2, \cdots, r_G\})}. \quad (3)$$

We give a comparison of GRPO and PPO in Supplementary Information, section 1.3.

Reward design

The reward is the source of the training signal, which decides the direction of RL optimization. For DeepSeek-R1-Zero, we use rule-based rewards to deliver precise feedback for data in mathematical, coding and logical reasoning domains. For DeepSeek-R1, we extend this approach by incorporating both rule-based rewards for reasoning-oriented data and model-based rewards for general data, thereby enhancing the adaptability of the learning process across diverse domains.

Rule-based rewards. Our rule-based reward system mainly consists of two types of reward: accuracy rewards and format rewards.

Accuracy rewards evaluate whether the response is correct. For example, in the case of maths problems with deterministic results, the model is required to provide the final answer in a specified format (for example, within a box), enabling reliable rule-based verification of correctness. Similarly, for code competition prompts, a compiler can be used to evaluate the responses of the model against a suite of predefined test cases, thereby generating objective feedback on correctness.

Format rewards complement the accuracy reward model by enforcing specific formatting requirements. In particular, the model is incentivized to encapsulate its reasoning process within designated tags, specifically <think> and </think>. This ensures that the thought process of the model is explicitly delineated, enhancing interpretability and facilitating subsequent analysis.

$$\text{Reward}_{\text{rule}} = \text{Reward}_{\text{acc}} + \text{Reward}_{\text{format}} \quad (4)$$

The accuracy, reward and format reward are combined with the same weight. Notably, we abstain from applying neural reward models—whether outcome-based or process-based-to reasoning tasks. This decision is predicated on our observation that neural reward models are susceptible to reward hacking during large-scale RL. Moreover, retraining such models necessitates substantial computational resources and introduces further complexity into the training pipeline, thereby complicating the overall optimization process.

Model-based rewards. For general data, we resort to reward models to capture human preferences in complex and nuanced scenarios. We build on the DeepSeek-V3 pipeline and use a similar distribution of preference pairs and training prompts. For helpfulness, we focus exclusively on the final summary, ensuring that the assessment emphasizes the use and relevance of the response to the user while minimizing interference with the underlying reasoning process. For harmlessness, we evaluate the entire response of the model, including both the reasoning process and the summary, to identify and mitigate any potential risks, biases or harmful content that may arise during the generation process.

Helpful reward model. For helpful reward model training, we first generate preference pairs by prompting DeepSeek-V3 using the Arena-Hard prompt format, listed in Supplementary Information, section 2.2, for which each pair consists of a user query along with two candidate responses. For each preference pair, we query DeepSeek-V3 four times, randomly assigning the responses as either Response A or Response B to mitigate positional bias. The final preference score is determined by averaging the four independent judgments, retaining only those pairs for which the score difference (Δ) exceeds 1 to ensure meaningful distinctions. Furthermore, to minimize length-related biases, we ensure that the chosen and rejected responses of the whole dataset have comparable lengths. In total, we curated 66,000 data pairs for training the reward model. The prompts used in this dataset are all non-reasoning questions and are sourced either from publicly available open-source datasets or from users who have explicitly consented to share their data for the purpose of model improvement. The architecture of our reward model is consistent with that of DeepSeek-R1, with the addition of a reward head designed to predict scalar preference scores.

$$\text{Reward}_{\text{helpful}} = \text{RM}_{\text{helpful}}(\text{Response}_A, \text{Response}_B) \quad (5)$$

The helpful reward models were trained with a batch size of 256, a learning rate of 6×10^{-6} and for a single epoch over the training dataset. The maximum sequence length during training is set to 8,192 tokens, whereas no explicit limit is imposed during reward model inference.

Safety reward model. To assess and improve model safety, we curated a dataset of 106,000 prompts with model-generated responses annotated as 'safe' or 'unsafe' according to predefined safety guidelines. Unlike the pairwise loss used in the helpfulness reward model, the safety reward model was trained using a pointwise methodology to distinguish between safe and unsafe responses. The training hyperparameters are the same as the helpful reward model.

$$\text{Reward}_{\text{safety}} = \text{RM}_{\text{safety}}(\text{Response}) \quad (6)$$

For general queries, each instance is categorized as belonging to either the safety dataset or the helpfulness dataset. The general reward, $\text{Reward}_{\text{general}}$, assigned to each query corresponds to the respective reward defined in the associated dataset.

Training details

Training details of DeepSeek-R1-Zero. To train DeepSeek-R1-Zero, we set the learning rate to 3×10^{-6}, the Kullback–Leibler (KL) coefficient to 0.001 and the sampling temperature to 1 for rollout. For each question, we sample 16 outputs with a maximum length of 32,768 tokens before the 8.2k step and 65,536 tokens afterward. As a result, both the performance and response length of DeepSeek-R1-Zero exhibit a substantial jump at the 8.2k step, with training continuing for a total of 10,400 steps, corresponding to 1.6 training epochs. Each training step consists of 32 unique questions, resulting in a training batch size of 512 per step. Every 400 steps, we replace the reference model with

方法

GRPO

GRPO[9] 是我们用于训练DeepSeek-R1-Zero和DeepSeek-R1的强化学习算法。它最初被提出是为了简化训练过程并减少近端策略优化（PPO）的资源消耗[31]，PPO广泛用于大语言模型的强化学习阶段[32]。GRPO的流程如扩展数据图2所示。

对于每个问题 q，GRPO 从旧策略中采样一组输出 $\{o_1, o_2, \ldots, o_G\}$，然后通过最大化以下目标来优化策略模型 π_θ：

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}[q \sim P(Q), \{o_i\}_{i=1}^{G} \sim \pi_{\theta_{\text{old}}}(O|q)]$$
$$\frac{1}{G}\sum_{i=1}^{G}\left(\min\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{\text{old}}}(o_i|q)}A_i,\ \text{clip}\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{\text{old}}}(o_i|q)}, 1-\epsilon, 1+\epsilon\right)A_i\right)\right. \quad (1)$$
$$\left. - \beta\mathbb{D}_{KL}(\pi_\theta||\pi_{\text{ref}})\right),$$

$$\mathbb{D}_{\text{KL}}(\pi_\theta||\pi_{\text{ref}}) = \frac{\pi_{\text{ref}}(o_i|q)}{\pi_\theta(o_i|q)} - \log\frac{\pi_{\text{ref}}(o_i|q)}{\pi_\theta(o_i|q)} - 1, \quad (2)$$

其中 π_{ref} 是参考策略，ϵ 和 β 是超参数，A_i是优势，使用一组奖励 $\{r_1, r_2, \ldots, r_G\}$ 计算得出，这些奖励对应于每组中的输出：

$$A_i = \frac{r_i - \text{mean}(\{r_1, r_2, \cdots, r_G\})}{\text{std}(\{r_1, r_2, \cdots, r_G\})}. \quad (3)$$

我们在补充信息第 1.3 节中给出了 GRPO 和 PPO 的比较。

奖励设计

奖励是训练信号的来源，它决定了RL优化的方向。对于DeepSeek-R1-Zero，我们使用基于规则的奖励，为数学、编码和逻辑推理领域的数据提供精确的反馈。对于DeepSeek-R1，我们扩展了这种方法，通过结合用于推理导向数据的基于规则的奖励和用于通用数据的基于模型的奖励，从而增强了学习过程在不同领域中的适应性。

基于规则的奖励。我们的基于规则的奖励系统主要由两种类型的奖励组成：准确性奖励和格式奖励。

准确性奖励评估响应是否正确。例如，在具有确定性结果的数学问题中，模型需要以指定格式（例如，在框中）提供最终答案，从而能够对正确性进行可靠的基于规则的验证。同样，对于代码竞赛提示，可以使用编译器根据一组预定义的测试用例来评估模型的响应，从而生成关于正确性的客观反馈。

格式奖励通过强制特定的格式要求来补充准确性奖励模型。特别是，模型被激励将其推理过程封装在指定的标签内，具体为 <think> and </think>. 这确保了模型的思维过程被明确 delineated，增强了可解释性并促进了后续分析。

$$\text{Reward}_{\text{rule}} = \text{Reward}_{\text{acc}} + \text{Reward}_{\text{format}} \quad (4)$$

准确率、奖励和格式奖励以相同权重进行组合。值得注意的是，我们避免对推理任务应用神经奖励模型——无论是基于结果的还是基于过程的。这一决策基于我们的观察：神经奖励模型在大规模强化学习过程中容易受到奖励攻击。

此外，重新训练此类模型需要大量计算资源，并给训练流程引入更多复杂性，从而使整体优化过程更加复杂。

基于模型的奖励。对于通用数据，我们借助奖励模型来捕捉复杂且细微场景中的人类偏好。我们基于DeepSeek-V3的流程构建，并使用类似分布的偏好对和训练提示。在有用性方面，我们仅关注最终摘要，确保评估侧重于响应对用户的使用价值和相关性，同时尽量减少对底层推理过程的干扰。在无害性方面，我们评估模型的完整响应，包括推理过程和摘要，以识别并缓解生成过程中可能出现的任何潜在风险、偏见或有害内容。

有用性奖励模型。对于有用性奖励模型的训练，我们首先使用Arena-Hard提示格式（列于补充信息第2.2节）提示DeepSeek-V3来生成偏好对，其中每一对包含一个用户查询和两个候选响应。对于每个偏好对，我们查询DeepSeek-V3四次，将响应随机分配为响应A或响应B以减轻位置偏差。最终偏好分数由四次独立判断的平均值确定，仅保留分数差（Δ）超过1的配对，以确保有意义的区分。此外，为了最小化长度相关偏差，我们确保整个数据集中被选中和被拒绝的响应长度相当。我们总共整理了66,000个数据对用于训练奖励模型。该数据集中使用的提示均为非推理问题，来源为公开可用的开源数据集或明确同意共享数据以用于模型改进的用户。我们的奖励模型架构与DeepSeek-R1一致，并增加了一个奖励头，用于预测标量偏好分数。

$$\text{Reward}_{\text{helpful}} = \text{RM}_{\text{helpful}}(\text{Response}_A, \text{Response}_B) \quad (5)$$

有用性奖励模型以256的批量大小、6×10^{-6} 的学习率，在训练数据集上训练了一个epoch。训练期间的最大序列长度设置为8,192个token，而奖励模型推理期间没有施加显式限制。

安全奖励模型。为了评估和改进模型安全性，我们整理了一个包含106,000个提示的数据集，其中模型生成的回复根据预定义的安全准则被标注为“安全”或“不安全”。与有用性奖励模型中使用的成对损失不同，安全奖励模型使用逐点方法进行训练，以区分安全和unsafe 回复。训练超参数与有用性奖励模型相同。

$$\text{Reward}_{\text{safety}} = \text{RM}_{\text{safety}}(\text{Response}) \quad (6)$$

对于一般查询，每个实例被归类为属于安全数据集或有用性数据集。分配给每个查询的一般奖励 $\text{Reward}_{\text{general}}$ 对应于相关数据集中定义的相应奖励。

训练细节

DeepSeek-R1-Zero 的训练细节。为了训练 DeepSeek-R1-Zero，我们将学习率设为 3×10^{-6}，Kullback–Leibler（KL）系数设为 0.001，采样温度设为 1 用于 rollout。对于每个问题，我们在 8.2k 步之前采样 16 个输出，最大长度为 32,768 个 token，之后为 65,536 个 token。因此，DeepSeek-R1-Zero 的性能和响应长度在 8.2k 步时都出现了显著跃升，训练总共持续 10,400 步，对应 1.6 个训练 epoch。每个训练步包含 32 个不同问题，从而导致每步的训练批次大小为 512。每 400 步，我们将参考模型替换为

the latest policy model. To accelerate training, each rollout generates 8,192 outputs, which are randomly split into 16 minibatches and trained for only a single inner epoch.

Training details of the first RL stage. In the first stage of RL, we set the learning rate to 3×10^{-6}, the KL coefficient to 0.001, the GRPO clip ratio ϵ to 10 and the sampling temperature to 1 for rollout. For each question, we sample 16 outputs with a maximum length of 32,768. Each training step consists of 32 unique questions, resulting in a training batch size of 512 per step. Every 400 steps, we replace the reference model with the latest policy model. To accelerate training, each rollout generates 8,192 outputs, which are randomly split into 16 minibatches and trained for only a single inner epoch. However, to mitigate the issue of language mixing, we introduce a language consistency reward during RL training, which is calculated as the proportion of target language words in the CoT.

$$\text{Reward}_{\text{language}} = \frac{\text{Num}(\text{Words}_{\text{target}})}{\text{Num}(\text{Words})} \tag{7}$$

Although ablation experiments in Supplementary Information, sction 2.6 show that such alignment results in a slight degradation in the performance of the model, this reward aligns with human preferences, making it more readable. We apply the language consistency reward to both reasoning and non-reasoning data by directly adding it to the final reward.

Note that the clip ratio plays a crucial role in training. A lower value can lead to the truncation of gradients for a large number of tokens, thereby degrading the performance of the model, whereas a higher value may cause instability during training. Details of RL data used in this stage are provided in Supplementary Information, section 2.3.

Training details of the second RL stage. Specifically, we train the model using a combination of reward signals and diverse prompt distributions. For reasoning data, we follow the methodology outlined in DeepSeek-R1-Zero, which uses rule-based rewards to guide learning in mathematical, coding and logical reasoning domains. During the training process, we observe that CoT often exhibits language mixing, particularly when RL prompts involve several languages. For general data, we use reward models to guide training. Ultimately, the integration of reward signals with diverse data distributions enables us to develop a model that not only excels in reasoning but also assigns priority to helpfulness and harmlessness. Given a batch of data, the reward can be formulated as

$$\text{Reward} = \text{Reward}_{\text{reasoning}} + \text{Reward}_{\text{general}} + \text{Reward}_{\text{language}} \tag{8}$$

in which

$$\text{Reward}_{\text{reasoning}} = \text{Reward}_{\text{rule}} \tag{9}$$

$$\text{Reward}_{\text{general}} = \text{Reward}_{\text{reward_model}} + \text{Reward}_{\text{format}} \tag{10}$$

The second stage of RL retains most of the parameters from the first stage, with the key difference being a reduced temperature of 0.7, as we find that higher temperatures in this stage lead to incoherent generation. The stage comprises a total of 1,700 training steps, during which general instruction data and preference-based rewards are incorporated exclusively in the final 400 steps. We find that more training steps with the model-based preference reward signal may lead to reward hacking, which is documented in Supplementary Information, section 2.5.

Data availability

We provide the data samples used in our rejection sampling and RL prompts at https://github.com/deepseek-ai/DeepSeek-R1 (https://doi.org/10.5281/zenodo.15753193)[33]. Comprehensive statistics and details of our complete data-generation methodology are presented in Supplementary Information, section 2.3.

Code availability

Trained weights of DeepSeek-R1-Zero and DeepSeek-R1 are available under an MIT license at https://github.com/deepseek-ai/DeepSeek-R1 (https://doi.org/10.5281/zenodo.15753193)[33]. The inference script is released at https://github.com/deepseek-ai/DeepSeek-V3 (https://doi.org/10.5281/zenodo.15753347)[34]. Neural networks were developed with PyTorch[35] and the distributed framework is based on our internal framework HAI-LLM (https://www.high-flyer.cn/en/blog/hai-llm). The inference framework is based on vLLM[36]. Data analysis used Python v.3.8 (https://www.python.org/), NumPy v.1.23.1 (https://github.com/numpy/numpy), Matplotlib v.3.5.2 (https://github.com/matplotlib/matplotlib) and TensorBoard v.2.9.1 (https://github.com/tensorflow/tensorboard).

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最新的策略模型。为了加速训练，每次 rollout 生成 8,192 个输出，这些输出被随机分成 16 个 minibatch，并且仅训练一个内部 epoch。

首个RL阶段的训练细节。在RL的第一阶段，我们将学习率设为3×10^{-6}，KL系数设为0.001，GRPO裁剪ratioϵto设为10，并将rollout的采样温度设为1。对于每个问题，我们采样16个输出，最大长度为32,768。每个训练步包含32个不同的问题，因此每步的训练批量大小为512。每400步，我们将参考模型替换为最新的策略模型。为加速训练，每次rollout生成8,192个输出，这些输出被随机分为16个小批量，并且仅训练一个内部epoch。然而，为缓解语言混合问题，我们在RL训练期间引入了语言一致性奖励，该奖励计算为CoT中目标语言词汇的比例。

$$\text{Reward}_{\text{language}} = \frac{\text{Num}(\text{Words}_{\text{target}})}{\text{Num}(\text{Words})} \tag{7}$$

尽管补充信息第2.6节中的消融实验表明，这种对齐会导致模型性能略有下降，但该奖励与人类偏好一致，使其更具可读性。我们通过直接将语言一致性奖励添加到最终奖励中，将其应用于推理和非推理数据。

注意，裁剪比例在训练中起着至关重要的作用。较低的值可能导致大量token的梯度被截断，从而降低模型的性能，而较高的值则可能导致训练过程中的不稳定性。此阶段所用强化学习数据的详细信息见补充信息第2.3节。

第二个强化学习阶段的训练细节。具体而言，我们使用奖励信号和多样化提示分布的组 合来训练模型。对于推理数据，我们遵循DeepSeek-R1-Zero中概述的方法，该方法使用基于规则的奖励来指导数学、编码和逻辑推理领域的学习。在训练过程中，我们观察到CoT经常表现出语言混合，尤其是当强化学习提示涉及多种语言时。对于通用数据，我们使用奖励模型来指导训练。最终，奖励信号与多样化数据分布的整合使我们能够开发出一个不仅在推理方面表现出色，而且优先考虑有用性和无害性的模型。给定一批数据，奖励可以表述为

$$\text{Reward} = \text{Reward}_{\text{reasoning}} + \text{Reward}_{\text{general}} + \text{Reward}_{\text{language}} \tag{8}$$

其中

$$\text{Reward}_{\text{reasoning}} = \text{Reward}_{\text{rule}} \tag{9}$$

$$\text{Reward}_{\text{general}} = \text{Reward}_{\text{reward_model}} + \text{Reward}_{\text{format}} \tag{10}$$

强化学习的第二阶段保留了第一阶段的大部分参数，关键区别在于温度降低至0.7，因为我们发现在此阶段使用更高的温度会导致生成不连贯。该阶段总共包含1,700个训练步骤，其中通用指令数据和基于偏好的奖励仅在最后400个步骤中引入。我们发现，更多使用基于模型的偏好奖励信号进行训练步骤可能导致奖励黑客行为，这在补充信息第2.5节中有记录。

数据可用性

我们在以下地址提供了用于拒绝采样和强化学习提示的数据样本：https://github.com/deepseek-ai/DeepSeek-R1（https://doi.org/10.5281/zenodo.15753193）[33]。完整的统计数据以及我们完整数据生成方法的详细信息见补充信息第2.3节。

代码可用性

DeepSeek-R1-Zero和DeepSeek-R1的训练权重以MIT许可协议提供，地址为https://github.com/deepseek-ai/DeepSeek-R1（https://doi.org/10.5281/zenodo.15753193）[33]。推理脚本发布于https://github.com/deepseek-ai/DeepSeek-V3（https://doi.org/10.5281/zenodo.15753347）[34]。神经网络使用PyTorch开发[35]，分布式框架基于我们的内部框架HAI-LLM（https://www.high-flyer.cn/en/blog/hai-llm）。推理框架基于vLLM[36]。数据分析使用了Python v.3.8（https://www.python.org/）、NumPy v.1.23.1（https://github.com/numpy/numpy）、Matplotlib v.3.5.2（https://github.com/matplotlib/matplotlib）和TensorBoard v.2.9.1（https://github.com/tensorflow/tensorboard）。

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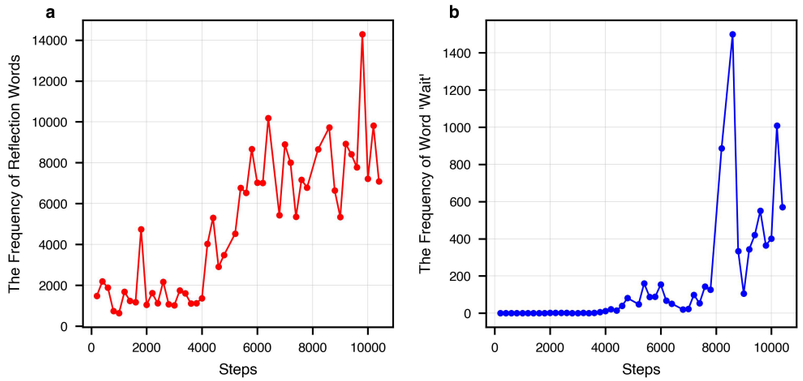
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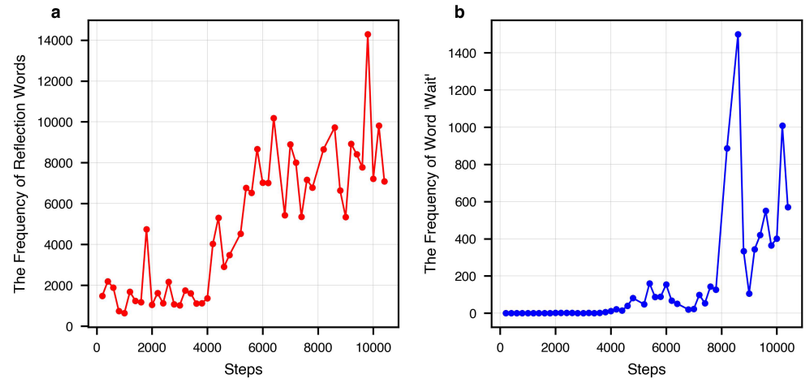
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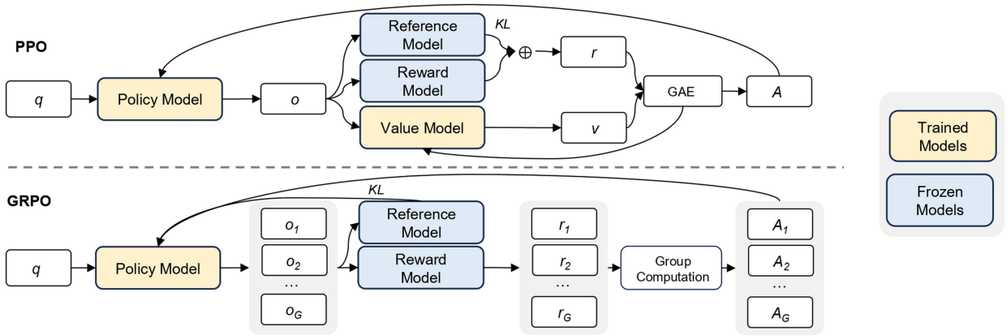
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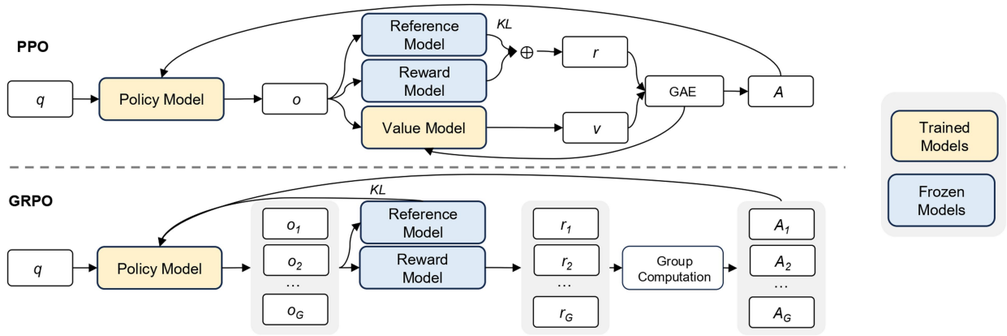
Extended Data Fig. 1 | Evolution of reasoning-related linguistic features in model outputs across training steps. a, Frequency of representative reflective terms in model-generated outputs throughout the training process. Reflective terms—including 'wait', 'mistake', 'however', 'but', 'retry', 'error', 'verify', 'wrong', 'evaluate' and 'check'—were identified and curated by a panel of three human experts. Each expert independently proposed a set of words indicative of reflective reasoning, which were subsequently consolidated through consensus into a final vocabulary list. **b**, Frequency of the term 'wait' in model outputs over the course of training. This term was virtually absent during the initial training stages, appeared sporadically between steps 4,000 and 7,000 and exhibited a marked increase in frequency after step 8,000. These trends suggest the emergence of temporal reasoning or self-monitoring behaviour as training progresses.



扩展数据图 1 | 模型输出中与推理相关的语言特征在训练步骤中的演变。 a，模型生成输出中代表性反思性词汇在整个训练过程中的出现频率。反思性词汇——包括“wait”、“mistake”、“however”、“but”、“retry”、“error”、“verify”、“wrong”、“evaluate”和“check”——由三名人类专家组成的小组进行识别和整理。每位专家独立提出一组指示反思性推理的词汇，随后这些词汇通过共识整合为最终词汇列表。 **b**，模型输出中“wait”一词在训练过程中的出现频率。该词在初始训练阶段几乎不存在，在第4,000步至7,000步之间零星出现，并在第8,000步之后出现频率显著增加。这些趋势表明，随着训练的推进，时间推理或自我监控行为逐渐涌现。



Extended Data Fig. 2 | Illustration of the proposed GRPO for RL-based training. In the proposed framework, a LLM is used as a policy model to generate responses $\{o_1, o_2, \ldots, o_G\}$ conditioned on a given query q. Each response within the group is evaluated by a reward model—either learned (model-based) or manually specified (rule-based)—to assign a scalar reward signal. Subsequently, GRPO computes the relative advantages of each group member based on their assigned rewards. Rather than relying on an explicit value function, as in PPO, GRPO directly estimates advantages from the intra-group reward distribution. The policy parameters are then updated to maximize the expected reward while simultaneously minimizing divergence from a reference policy, typically quantified through the KL divergence. By eliminating the need for a separate value network, GRPO offers a simplified yet effective alternative to traditional actor-critic methods such as PPO.



扩展数据图 2 | 所提出的用于基于强化学习的GRPO的图示
训练。在所提出的框架中，LLM被用作策略模型，以在给定查询q的条件下生成响应$\{o_1, o_2, \ldots, o_G\}$。组内的每个响应由奖励模型——无论是学习得到的（基于模型）还是手动指定的（基于规则）——进行评估，以分配标量奖励信号。随后，GRPO根据分配的奖励计算每个组成员的相对优势。与PPO中依赖显式价值函数不同，GRPO直接从组内奖励分布估计优势。然后更新策略参数，以最大化期望奖励，同时最小化与参考策略的散度，通常通过KL散度来量化。通过消除对单独价值网络的需求，GRPO为传统演员-评论家方法（如PPO）提供了一种简化而有效的替代方案。

Extended Data Table 1 | An illustrative example from the AIME dataset

PROMPT

Let $b \geq 2$ be an integer. Call a positive integer n b-*eautiful* if it has exactly two digits when expressed in base b, and these two digits sum to $\sqrt{n}$. For example, 81 is 13-eautiful because $81 = \underline{6}\,\underline{3}_{13}$ and $6 + 3 = \sqrt{81}$. Find the least integer $b \geq 2$ for which there are more than ten b-eautiful integers.

Please reason step by step, and put your final answer within \boxed{}.

Evaluation

Parse the final answer within \boxed{} and use a rule-based grader to determine if it equals the ground truth. Round numerical values as needed, and use 'SymPy' to parse expressions.

The presented question is sourced from the 2024 AIME. The model is tasked with solving the problem and formatting its answer in a required format (for example, ANSWER). For evaluation, a rule-based grading system is used to determine correctness. The output of the model is considered correct if and only if it exactly matches the ground-truth solution.

扩展数据表 1 | 来自AIME数据集的一个示例

PROMPT

Let $b \geq 2$ be an integer. Call a positive integer n b-*eautiful* if it has exactly two digits when expressed in base b, and these two digits sum to $\sqrt{n}$. For example, 81 is 13-eautiful because $81 = \underline{6}\,\underline{3}_{13}$ and $6 + 3 = \sqrt{81}$. Find the least integer $b \geq 2$ for which there are more than ten b-eautiful integers.

Please reason step by step, and put your final answer within \boxed{}.

Evaluation

Parse the final answer within \boxed{} and use a rule-based grader to determine if it equals the ground truth. Round numerical values as needed, and use 'SymPy' to parse expressions.

所呈现的问题源自2024年AIME。模型的任务是解决该问题并按照要求的格式（例如，ANSWER）格式化其答案。在评估时，使用基于规则的评分系统来确定正确性。当且仅当模型的输出与真实解决方案完全匹配时，该输出才被视为正确。